# Amit Kumar Mohanty

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*Target Role: AI/ML Engineer - AI Ops (Predictive ML + MLOps)*

## PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of experience designing, training, and deploying predictive models and owning MLOps pipelines end-to-end. Hands-on with anomaly detection, classification, and time-series style forecasting using the Python ML stack (scikit-learn, PyTorch, TensorFlow), with rigorous evaluation, feature engineering, and model monitoring. Experienced building agentic AI decision loops and LLM integrations, and serving models via low-latency FastAPI inference APIs on AWS and GCP. Strong experimental mindset, statistical rigor, and clear documentation of model architectures, datasets, and evaluation metrics.

## EDUCATION

**K.R. Mangalam University, Gurgaon** — **July 2021 - July 2025**

*B.Tech in Computer Science and Engineering (Specialization in AI/ML)*

**Birla Institute of Technology and Science, Pilani (BITS Pilani)** — **July 2026 - July 2028**

*M.Tech in AI & ML (Pursuing) - Coursework: Deep Learning, Deep Reinforcement Learning*

## EXPERIENCE

**Research Associate - AI, Keywords Studios India - Gurgaon, Haryana** — **May 2025 - Present**

- Designed agentic AI decision loops (detect-decide-act) for autonomous multi-step task execution, improving reliability and reducing manual intervention.
- Owned MLOps workflow end-to-end - data ingestion, feature engineering, training, validation, and deployment - delivering 85% of training data to production; awarded "Rookie of the Year 2025."
- Defined rigorous evaluation metrics and structured test reviews to improve model accuracy and monitor performance in production.

**Artificial Intelligence Intern, Infosys - Remote** — **Nov 2024 - Feb 2025**

- Built LLM integrations for knowledge assistance and runbook-style responses with LangChain, improving response accuracy by 35%.
- Implemented automated evaluation with confidence scoring that reduced errors by 20%, and optimized inference for latency (-25%) within a 12-engineer Agile team.

## PROJECTS

**Diabetic Optiscan - Classification & Anomaly Detection** — **Jan 2025 - June 2025**

- Built and fine-tuned a Vision Transformer (ViT-B/16) for high-precision image classification/anomaly detection; co-authored a peer-reviewed paper with rigorous evaluation metrics.
- Created a custom wavelet technique improving detection precision by 50%, avoiding overfitting via structured validation.

**Dynamic Screen Companion - Low-Latency Inference API** — **June 2025 - Present**

- Engineered a FastAPI inference backend serving concurrent real-time predictions, containerized with Docker + CI/CD for scalable, resilient deployment.
- Integrated LLMs with a retrieval/knowledge-indexing layer and monitored model performance on live signals.

## SKILLS

**Machine Learning & Statistics:** Supervised & Unsupervised ML, Anomaly Detection, Classification, Regression, Time-Series Forecasting, Model Evaluation, Overfitting Control, Statistical Analysis (Hypothesis Testing, Confidence Intervals, Distributions)

**Python ML Stack:** Python, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, NumPy, Pandas

**Time-Series & Deep Learning:** LSTM, Transformer-based Models, Prophet/ARIMA, CNNs, Vision Transformers, Feature Engineering (metrics, logs, events)

**MLOps:** MLflow, Weights & Biases, Model Registry, Experiment Tracking, Drift Detection, Model Monitoring, Docker, Kubernetes, CI/CD

**Agentic AI & LLMs:** Agentic AI / Autonomous Decision Loops, Reinforcement Learning (foundations), Multi-Agent Systems, LLM Integration, RAG, Knowledge Base Indexing

**Inference APIs & Cloud:** FastAPI Low-Latency Inference APIs, REST APIs, AWS (SageMaker), GCP (Vertex AI), Azure ML, GPU Training, Vector Databases (FAISS/Chroma)

## CERTIFICATIONS

- ISRO - AI/ML for Geodata Analysis (time-series & spatial pattern analysis on remote-sensing data).
- Peer-reviewed research publication (Vision Transformers for medical image classification).
- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.